

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Ca-45			FORMS: Soluble		
PHYSICAL CHARACTERISTICS HALF-LIFE: 162.67 days DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
		77 100			
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY C – level (low hazard): ≤ 20 mCi B – level (moderate hazard): > 20 mCi , ≤ 1 Ci A – level (high hazard): > 1 Ci					
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: 0 mrem/h Beta dose rate to skin, point source at 1 ft, 1 mCi: 0 mrem/h Contamination skin dose, uniform deposit of 1 μ Ci per cm ² : 3100 mrem/h			INTERNAL RADIATION HAZARDS Annual Limit on Intake: 2000 μCi (Ingestion) 800 μCi (Inhalation) The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).		
SHIELDING Gammas/X-rays: N/A Betas/electrons: <1 mm of plastic will absorb all emissions.			DOSIMETRY AND BIOASSAY REQs Urine assays may be required after large spills or contaminations.		
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Recommended survey probe: PGM (low efficiency, smear surveys are optimal) 2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 3. Survey work areas before, during, and after work. Smear surveys are required. 4. Segregate waste to those with half-lives of greater than 120 days (excluding H3 and/or C14). 5. Limit soluble waste to the sewer to less than 100 μCi/day per lab.					

References:

- Delacroix, D., Guerre, J.P., Leblanc, P., Hickman, C. (2002). Radionuclide and Radiation Protection Data Handbook (2nd ed.). Ashford, Kent: Nuclear Technology Publishing.
- Johnson, T.E., Birk, B.K. (2012). Health Physics and Radiological Health (4th ed.). Baltimore, MD: Lippincott Williams & Wilkins.
- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>